the remaining three places can be filled with remaining 4 digits in ${}^4P_3 = 24$ ways. This means, the number of 4-digit numbers having 1 in units place is ${}^4P_3 = 24$. Similarly, each of the digits 2, 4, 6, 8 appear 24 times in units place. An addition of all these digits gives the sum of all the unit digits of all 120 numbers. Therefore,

$$({}^4P_3 \times 1) + ({}^4P_3 \times 2) + ({}^4P_3 \times 4) + ({}^4P_3 \times 6) + ({}^4P_3 \times 8)$$

$$\begin{aligned} &= {}^4P_3 \times (1 + 2 + 4 + 6 + 8) \\ &= {}^4P_3 \times (\text{sum of the digits}) \\ &= {}^4P_3 \times 21. \end{aligned}$$

Similarly, we get the sum of the digits in 10th place as ${}^4P_3 \times 21$. Since it is in 10th place, its value is ${}^4P_3 \times 21 \times 10$. Similarly, the values of the sum of the digits in 100th place and 1000^{th}place are ${}^4P_3 \times 21 \times 100$ and ${}^4P_3 \times 21 \times 1000$ respectively. Hence the sum of all the 4 digit numbers formed by using the digits 1, 2, 4, 6, 8 is

$$({}^4P_3 \times 21) + ({}^4P_3 \times 21 \times 10) + ({}^4P_3 \times 21 \times 100) + ({}^4P_3 \times 21 \times 1000)$$

$$\begin{aligned} &= {}^4P_3\,(21 \times 1111) \\ &= 24 \times 21 \times 1111 \\ &= 559944. \end{aligned}$$

Deduction 4.1: The sum of all r-digit numbers that can be formed using the given n non zero digits is ${}^{(n-1)}P_{(r-1)} \times$ (sum of the digits) $\times\, 111\cdots1$(r times)

Deduction 4.2: If 0 is one digit among the given n digits, then we get that the sum of the r-digits numbers that can be formed using the given n digits (including 0) is
$\{{}^{(n-1)}P_{(r-1)} \times$ (sum of the digits) $\times\, 111\cdots1$(r times)$\}$ $-$
$\{{}^{(n-2)}P_{(r-2)} \times$ (sum of the digits) $\times\, 111\cdots1$((r-1) times)$\}$.

Permutation as Function

Permutation on any finite set $S_n = \{x_1, x_2, x_3, \cdots, x_n\}$ is a bijective function from $S_n \to S_n$. Therefore the set of all permutations on a finite set with n elements is the same as the total number of bijective functions on the set. This is precisely $n!$. Hence the study of permutation is the same as the study of the bijective mappings on the set. Few representations for a permutation on S_3 are given by

$$\begin{bmatrix} A & B & C \\ B & C & A \end{bmatrix}, \begin{bmatrix} A & B & C \\ C & A & B \end{bmatrix}, \begin{bmatrix} A & B & C \\ C & B & A \end{bmatrix}, \cdots$$



Exercise - 4.2

1. If ${}^{(n-1)}P_3 :^n P_4 = 1 : 10$, find n.
2. If ${}^{10}P_{r-1} = 2 \times^6 P_r$, find r.
3. (i) Suppose 8 people enter an event in a swimming meet. In how many ways could the gold, silver and bronze prizes be awarded?
   (ii) Three men have 4 coats, 5 waist coats and 6 caps. In how many ways can they wear them?